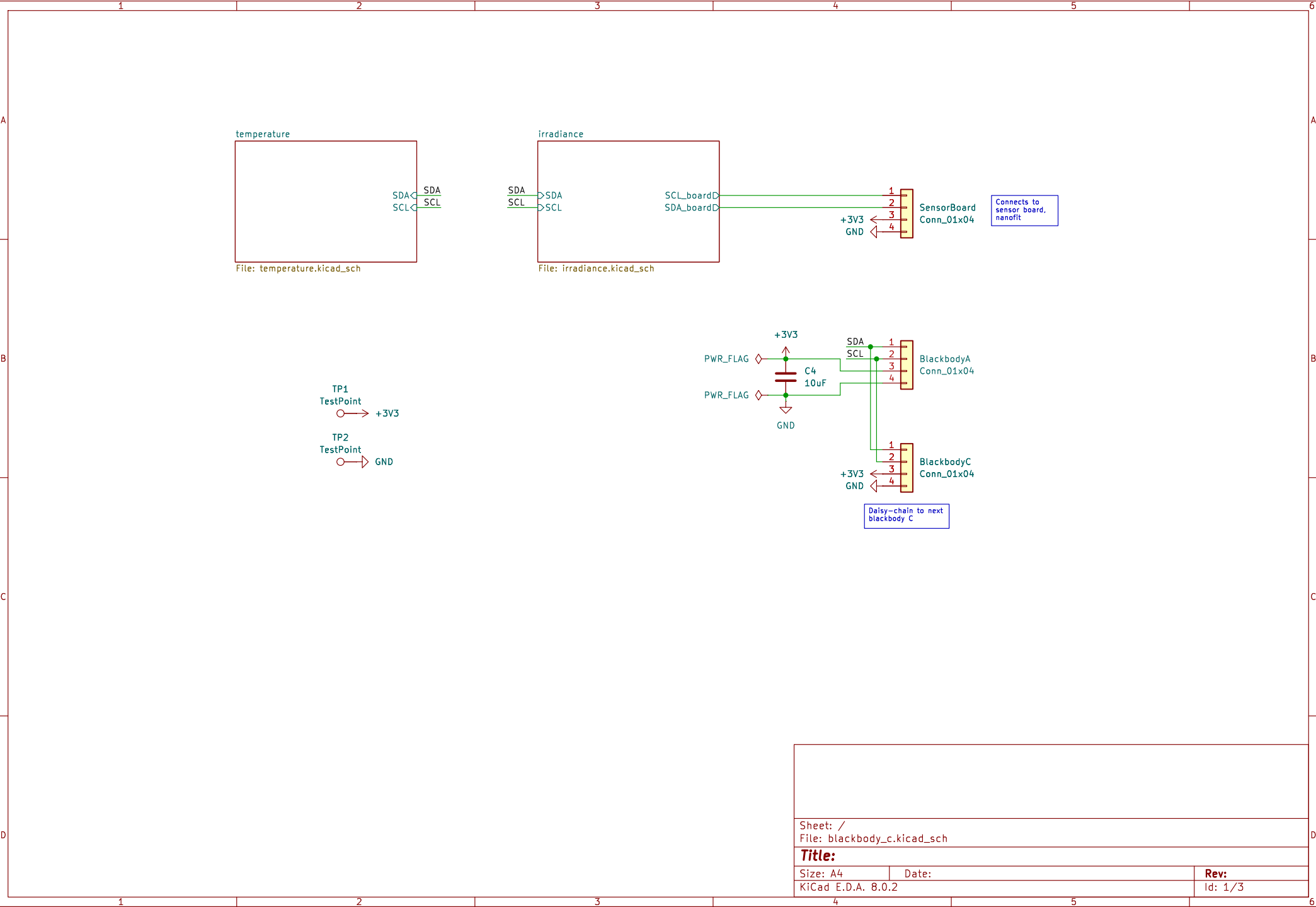






6.3.3.1 Open-Circuit Detection Technique

For open-circuit detection, the Input Range Flag bit, bit 4 of the STATUS register (Register 5-6), can be used to detect open-circuit conditions. This would require a few external resistors, as shown in Figure 6-6. The passive circuit does not affect the MCP9600/L00/RL00 accuracy (the recommended value for R_B is set to 1 M Ω). When the thermocouple is connected, the input Common-mode voltage is $0.5 \cdot V_{DD}$. When the thermocouple is disconnected, the voltage at the V_{IN+} input is $0.66 \cdot V_{DD}$ and the voltage at the V_{IN-} input is pulled down to V_{SS} . This change forces the Input Range Flag bit to be set. The master controller can momentarily poll the Status bit to detect the open-circuit condition.

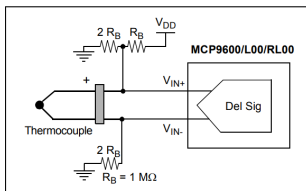
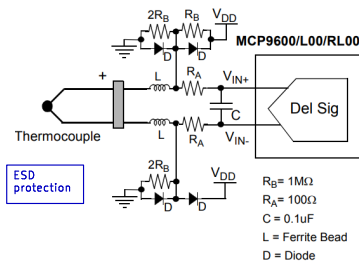
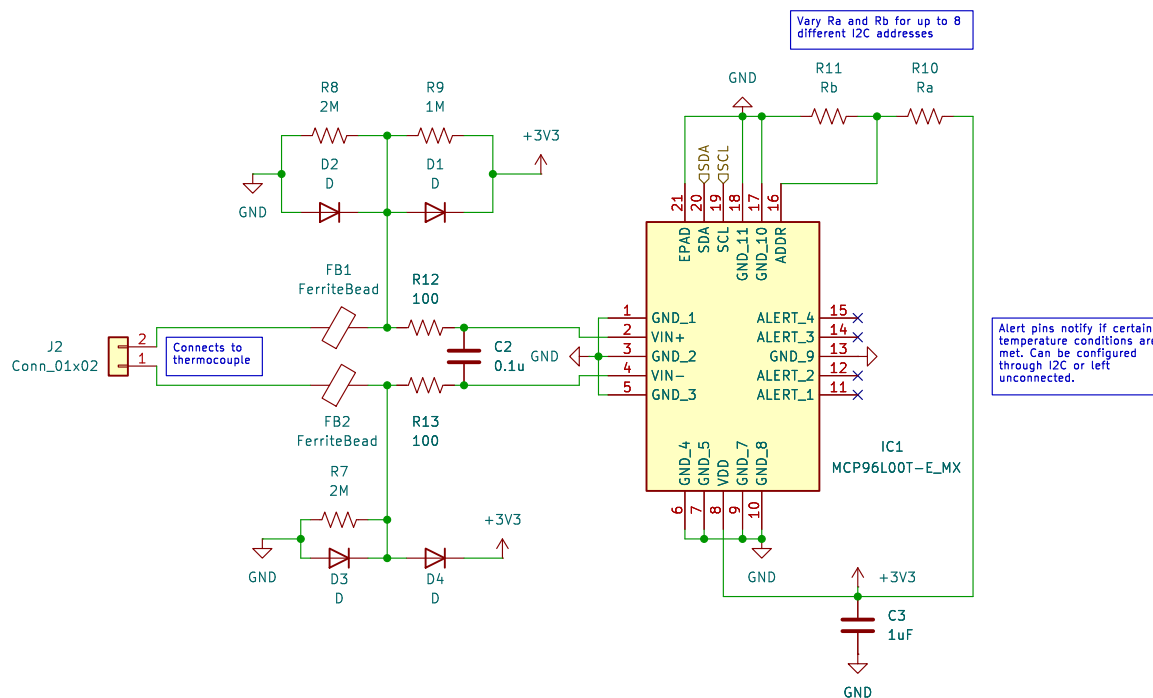


FIGURE 6-6: Adding Open-Circuit Detection Resistors.



Sheet: /temperature/
File: temperature.kicad_sch

Title:

Size: A4
KiCad E.D.A. 8.0.2

Date:

Rev:

Id: 3/3